

Tyler Jacob Tennant

Senior AI & Data Scientist | LLM & Generative AI Expert | Applied ML Leader
+1 (475) 739-1540 | tylerent517@gmail.com | Nevada, United States

PROFILE

Visionary and hands-on Senior AI/Data Scientist with 8+ years of experience delivering advanced machine learning, NLP, and Generative AI solutions in high-impact environments across robotics, cloud infrastructure, open-source AI, and consumer applications. Adept at designing and scaling LLMs, building cloud-native ML architectures, and driving cross-functional AI initiatives aligned with product and business strategy.

Proven track record in bridging the gap between ML research and production engineering, with expertise spanning multimodal learning, MLOps, data-centric AI, and real-time inference systems. Trusted technical leader, mentor, and builder of intelligent products used by millions.

PROFESSIONAL EXPERIENCE

Neusis

Senior Data Scientist | Generative AI & Multimodal Systems Lead
Aug 2023 – Present | Remote

- Led the architecture and deployment of foundation-model-powered systems combining **image recognition**, **LLM-based text generation**, and **user personalization** for collectible asset discovery and pricing.
- Built a visual-semantic embedding pipeline using **CLIP**, **custom ResNet encoders**, and **Scikit-learn** for feature-based similarity, boosting card match accuracy by 40%.
- Developed an **LLM-powered assistant** using **LangChain**, **OpenAI APIs**, and **Retrieval-Augmented Generation (RAG)**.
- Operationalized end-to-end ML pipelines with **Kubernetes**, **Docker**, and **GitHub Actions** for fast, reliable deployment.
- Used **PyTorch** and **TensorFlow** for training deep learning models; integrated **ONNX** for optimized inference.
- Partnered cross-functionally with product and marketing teams, contributing to a 32% boost in retention and 18% revenue growth.
- Mentored junior engineers on MLOps tooling (**MLflow**, **Weights & Biases**) and model validation with **XGBoost** and **LightGBM**.

Hugging Face

Data Scientist | Applied Research & Developer Tools
June 2021 – July 2023 | Remote

- Contributed to the **Transformers**, **Datasets**, and **Accelerate** libraries, optimizing training on **TPUs/GPUs** using **PyTorch** and **TensorFlow** backends.
- Created tutorials for fine-tuning **LLMs** using **PEFT**, **quantization**, and **ONNX Runtime** – adopted by thousands of users.

- Designed benchmark pipelines comparing **Scikit-learn**, **CatBoost**, and Hugging Face Transformers on classification and regression tasks.
- Led client implementations of **domain-specific LLMs**, deploying with **FastAPI**, **SageMaker**, and **Triton Inference Server**.
- Acted as a key technical advisor to Fortune 500 teams on **RAG architectures** and **ML system integration**.

NVIDIA

Machine Learning Engineer | Edge AI & SDKs
Jan 2020 – May 2021 | Santa Clara, CA

- Built real-time AI inference systems using **TensorRT**, **DeepStream**, and **Triton**, running models built with **TensorFlow**, **PyTorch**, and **ONNX**.
- Designed scalable training pipelines using **DDP**, **NCCL**, and models like **YOLOv5**, **ResNet**, **BERT** – trained with **XGBoost** for ensemble meta-learning.
- Worked on the **TAO Toolkit** team, simplifying training for vision models with **Keras** and **PyTorch Lightning** wrappers.
- Integrated ML modules in smart city infrastructure, including license plate recognition and behavioral event alerts.
- Presented at **GTC** on optimization strategies using **quantization** and **model pruning**.

Vicarious AI

Data Scientist | Robotics & Cognitive AI
July 2017 – Dec 2019 | San Francisco, CA

- Developed **Recursive Cortical Networks (RCNs)** and **probabilistic graphical models** for robotic perception.
- Used **Scikit-learn** and **XGBoost** to validate perception outputs with traditional classifiers during simulation testing.
- Created attention-based spatial models using **TensorFlow** and **Keras** for intelligent robotic interaction with complex environments.
- Conducted feasibility studies on low-data regimes using **CatBoost** and **LightGBM** to evaluate edge deployment performance.
- Authored whitepapers that contributed to research now used in **Intrinsic's AI robotics platform**.

and trustworthiness of robotic actions.

EDUCATION

University of California, Berkeley
Bachelor of Science in Computer Science | 2013 – 2017

TECHNICAL EXPERTISE

Languages

Python, C++, Rust, JavaScript, TypeScript, Bash, SQL, R

Machine Learning & Deep Learning

LLMs, Transformers, Scikit-learn, XGBoost, Tensorflow, PyTorch, CNNs, RAG, PEFT, Quantization

NLP & Generative AI

Hugging Face Transformers, GPT-3/4, LLaMA, T5, BERT, LangChain, OpenAI API, Tokenization

MLOps & Deployment

MLflow, Kubeflow, Weights & Biases, Triton Inference Server, DVC, Jenkins, GitHub Actions, Docker, Kubernetes, CI/CD, Terraform

Cloud & Data Platforms

AWS (SageMaker, Lambda, EKS, Fargate), GCP (Vertex AI, GKE), Azure (AKS, ADF), Snowflake, BigQuery, S3, Redshift, Postgres, MongoDB

Data Engineering & ETL

Apache Airflow, Spark, Pandas, NumPy, SQLAlchemy, Data Lakes, dbt, Kafka

Frontend (for ML Interfaces)

React.js, Next.js, Angular, Tailwind CSS, Chart.js, Material UI

PROJECT HIGHLIGHTS

- **Multimodal LLM Chatbot for Collectibles:** Combined CLIP + LLaMA + LangChain with vector search for real-time Q&A and card valuation. Deployed via FastAPI and Sagemaker.
- **Enterprise RAG Toolkit:** Delivered plug-and-play retrieval-augmented generation systems using open-source LLMs + custom data pipelines, boosting response accuracy by 28%.
- **Smart City AI Stack:** Deployed scalable real-time inference with DeepStream + PyTorch for object tracking, alert generation, and analytics dashboards.
- **Open-Source ML Advocacy:** Spoke at PyData, GTC, and HF Meetups on topics including reproducibility, model compression, and LLM fine-tuning workflows.

LEADERSHIP & STRATEGY

- Led cross-functional teams of 5–12 engineers on AI-first product initiatives, including ML experimentation, deployment, and productization.
- Developed documentation and onboarding systems for data science teams to accelerate experimentation and reduce time-to-delivery by 40%.
- Drove roadmap alignment between research and product teams, translating abstract AI capabilities into user-facing applications.